# Paleomagnetic results from three rock units of New Brunswick and their bearing on the Lower Paleozoic tectonics of North America[1]

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Thermal, alternating field (AF), and two-stage (AF followed by thermal) experiments performed on one Silurian (Mascarene Group) and two Devonian (St. George and St. Stephen plutons) rock units indicate that considerable polar movement relative to New Brunswick took place in Siluro-Devonian times. The Mascarene Group yields a pole at 122°E, 02°S, the St. Stephen pluton, a pole at 136°E, 43°S, and the St. George pluton, two poles at 085°E, 38°S and 093°E, 29°N respectively. Although the southern poles lie some 70° south of most North American Silurian and Devonian poles, they are in agreement with poles obtained from two recent studies of Newfoundland rock units. It is unknown if this apparent polar displacement is representative of the entire North American craton or of its eastern margin only. A graphic representation of all available North American data is disconcerting and shows that little is known about the behaviour of the Lower Paleozoic field relative to North America and that many more data are needed to reconstruct the paleomagnetic record.

The St. George pluton carries two dual polarity remanences which often are not easily discernible, especially since one of the remanences has an intensity much weaker than the other. The usefulness of vector diagrams for the analysis of multicomponent remanences is demonstrated. Examples are shown where the dual polarity of the weaker remanence, which might otherwise escape notice, can be readily detected on enlarged scale vector diagrams.

Des traitements chimiques, thermiques et par champs alternatifs, en plus d'une méthode de désaimantation à deux étapes ont été appliqués sur des échantillons du groupe de Mascarene d'âge silurien et des plutons de St. George et de St. Stephen d'âge dévonien. Les résultats démontrent qu'il y a eu un déplacement considérable du pôle par rapport au Nouveau-Brunswick pendant la période siluro-dévonienne. Du groupe de Mascarene on obtient un pôle à 122°E, 02°S, du pluton de St. Stephen un pôle à 136°E, 43°S, et du pluton de St. George deux pôles, l'un à 085°E, 38°S et l'autre à 093°E, 29°N. Même si les deux pôles se situant près de la latitude 40°S sont 70° au sud de la majorité des pôles du Siluro-Dévonien de l'Amérique du Nord, ils sont corroborés par des pôles obtenus récemment de roches contemporaines provenant de Terre-Neuve. Présentement, il est impossible de savoir si ce déplacement rapide du pôle apparent doit être limité à la bordure continentale ou si on peut l'appliquer au continent nord-américain en entier. Une représentation graphique de toutes les données nord-américaines démontre que les particularités du champ magnétique en ce qui a trait à ce continent pour le Paléozoïque inférieur sont méconnues et que l'on a besoin de bien d'autres données afin de dévoiler l'enregistrement paléomagnétique de cette époque. Dans le pluton de St. George deux rémanences de polarité double sont difficilement séparables, d'autant plus que l'une a une intensité beaucoup plus faible que l'autre. On démontre l'utilité de diagrammes de vecteurs pour l'analyse des rémanences à composantes multiples en se servant d'exemples où la double polarité de la remanence faible, qui autrement pourrait passer inaperçue, se manifeste sur les diagrammes à échelle agrandie.

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## Introduction

The most deficient part of the Phanerozoic paleomagnetic record is the Lower Paleozoic. This is especially true for North America where the Lower Paleozoic (and here including the Devonian) results account for only 14% of the 291 Phanerozoic results as listed by Irving *et al.* (1976). As the Cambrian to Devonian interval covers 40% of the Phanerozoic time scale, the paleomagnetic data are therefore very unevenly distributed, averaging less than two data points per 10 Ma during this interval as compared to more than seven data points per 10 Ma for the more recent part of the Phanerozoic. Thus, it is not surprising that so little is known about the Lower Paleozoic magnetic field.

The Cambrian to Devonian data are shown in Fig. 1 and listed in Addendum A. Most of the poles fall into a wide swath running roughly from 190°E, 35°S to 110°E, 45°N, and, at first sight, this might give the impression that the Lower Paleozoic polar path for North America is fairly well defined and simple. However, it should be noted that the poles

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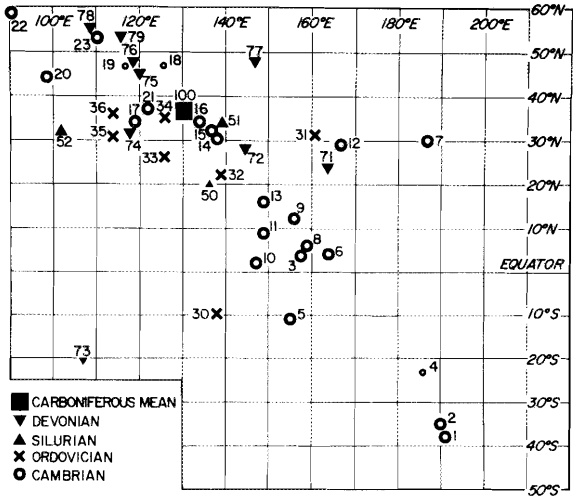



FIG. 1. Lower Paleozoic poles up to 1977. Related data and authorships are given in Addendum A. The symbols indicate the age of the rock unit which is not necessarily the age of the remanence from which they were derived. Small symbols indicate poles obtained from NRM (natural remanent magnetization), that is, no cleaning treatment was applied. The mean Carboniferous pole (100) is from Roy (1969). Data given in abstract form only are not included.

have been symbolized according to the best estimate of the age of the rock unit which is not necessarily the age of the magnetization vector from which the pole was obtained (pole age). The age of certain of these poles has been the subject of controversies. For example, Al-Khafaji and Vincenz (1971), Spall (1968), and Vincenz *et al.* (1975) have respectively suggested that the Lamotte result (pole 15), and the Wichita results (pole 17) and poles 14 and 16 obtained from Cambrian rock units represent a remagnetization during the Carboniferous because of the proximity of those poles with the Carboniferous pole (Fig. 1). The remagnetization hypothesis has been rejected by McElhinny and Opdyke (1973) on the basis that other Cambro-Ordovician (poles 20 and 21) and Ordovician (pole 36) results indicate that the Ordovician pole was near the Carboniferous pole. It has also been pointed out by Van der Voo *et al.* (1976) and French *et al.* (1977) that Cambrian rock units often carry multivectorial magnetizations which may indicate that appreciable apparent polar wander occurred during the Cambrian. The data shown in Fig. 1 tend to support this view since Cambrian poles are strung the full length of the swath.

Conversely, it is interesting to note that most of the Ordovician to Devonian poles lie near the Carboniferous pole. Does this mean that the apparent pole remained almost stationary for some 200 Ma (Ordovician to Carboniferous)? Or is it that some movement has remained undetected because of the limited number of data points? What is the significance of the few poles that do not fall into the general pattern? Clearly all of these questions cannot be answered satisfactorily with a single study of a few rock units. Many more data are needed. The purpose of this study is merely to contribute a few more data points and hopefully to elucidate some of these queries. The results are based on a limited number of sites, but the extensive experimental work performed seems to indicate that the Lower Paleozoic picture is much more complex than the simple one that could be inferred from the presently limited data base (Fig. 1). Many specimens carry two distinct remanences; each or both often having dual polarity. Vector diagrams such as those used by Roy and Park (1974) were found to be most useful for the recognition and analysis of these remanences. Enlarging the scale near the origin did provide one with much information about the weaker of the two remanences. Diagrams describing the changes of both the horizontal component and, most importantly, the total intensity vectors are shown to be a valuable visual aid for detailed vector analysis.

## Geology and Sampling

Lower Paleozoic rocks underlie the Passamaquoddy Bay region of southwestern New Brunswick (Fig. 2; Table 1). Although the oldest rocks of the region are the Lower Ordovician Charlotte Group, most of the bedrock is formed of Silurian and Devonian rocks. The Silurian succession is represented in New Brunswick by the Mascarene Group which comprises several formations (Oak Bay, Quoddy, Waweig, Pembroke, and Eastport) (Cumming 1967; Pickerill 1976). The age limits of the Mascarene Group are not accurately defined. Approximate limits of 420–400 Ma are given by Pajari *et al.* (1974). The uppermost formation (Eastport) may, however, be lowermost Devonian (Gedinnian $\approx$ 392 Ma) in age (Pickerill and Pajari 1976). The group consists of acid and basic volcanic rocks and associated clastic sedimentary rocks: conglomerates, limestones, and fossiliferous shales.

The Devonian rocks are represented by the Lower Devonian St. Stephen and St. George plutons, and the Upper Devonian Perry Formation. There is particularly good agreement between K–Ar (385 $\pm$ 11 and 391 $\pm$ 22 Ma; Wanless *et al.* 1973) and Rb–Sr (394 $\pm$ 20 Ma; Pajari *et al.* 1974) radiometric dates for the St. George pluton. It has

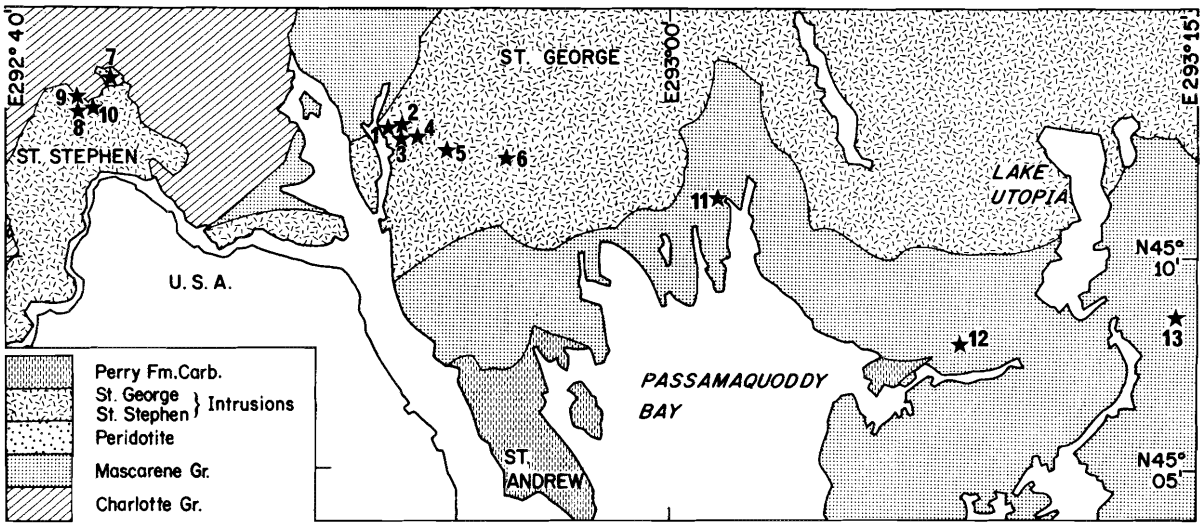



FIG. 2. Generalized map of southwestern New Brunswick, Canada, with sampling site localities.

TABLE 1. Summary of geological rock units and their relative radiometric ages

| Period | Time scale* (Ma) | Rock units† | Radiometric ages‡ (Ma) |
|---|---|---|---|
| | 345 | | 377–404 [5] |
| | | St. George | 385 ± 11 [4] |
| | | | 391 ± 22 [4] |
| | | | 394 ± 20 [3] |
| Devonian | | | |
| | | | 364 ± 16 [4] |
| | | St. Stephen | 350 ± 15 [4] |
| | | Ultrabasic | 401 ± 20 [4] |
| | | | 415 ± 19 [4] |
| | 395–405 | Eastport Fm. | 412 ± 5 [2] |
| | | Pembroke Fm. | 410 ± 15 [1] |
| Silurian | Mascarene Group | Waweig Fm. | 415 ± 15 [1] |
| | | Quoddy Fm. | 420 ± 15 [1] |
| | | Oak Bay Fm. | |
| | 430–440 | | |

*Combined Harland *et al.* (1964) and Kulp (1960).
†Modified from Pajari *et al.* (1974).
‡Source of radiometric data: [1] = Fullagar and Bottino, quoted in Pajari *et al.* (1974); [2] = Bottino and Fullagar (1966); [3] = Pajari *et al.* (1974); [4] = Wanless *et al.* (1973); [5] = Cumming (1967).

been suggested that this agreement rules out any appreciable updating by a thermal event (Pajari *et al.* 1974). Two samples from the St. Stephen pluton, originally believed to be Ordovician in age, have yielded K–Ar ages of 364 ± 16 and 350 ± 15 Ma from biotite and 401 ± 20 and 415 ± 19 Ma from coexisting hornblende respectively (Wanless *et al.* 1973). This has led to the suggestion that the St. Stephen pluton (average hornblende dates at 408 Ma) was emplaced shortly prior to the emplacement time of the St. George pluton and that the 357 Ma average age obtained from biotite may represent an updating due to a later thermal event (Pajari *et al.* 1974). In summary, Pajari *et al.* (1974) have concluded that "the last phase of the Acadian deformation in the area occurred between 394 and 410 Ma" (p. 1312), and regard the Mascarene Group as a pre-Acadian unit and the St. Stephen and St. George plutons as post-Acadian intrusions.

Six sites (1–6) were sampled from the St. George pluton (Fig. 2). The rocks are gabbroic in composition, except for site 2 where blocks of diorite are embedded into a monzonite matrix. Polished section examination reveals the presence of both hematite and magnetite at all sites, except site 5. Although hematite and magnetite are the major

opaque minerals, minor ilmenite and pyrite are also present. There is a mutual boundary between the hematite and magnetite, so that no age relationship can be defined. At site 5 the principal opaque mineral is ilmenite, seen in biotite as skeletal relicts. Four sites (7–10) were obtained from the St. Stephen pluton (Fig. 2). The rocks are gabbros except for site 7 which was sampled from a small peridotite body associated with the pluton. Both magnetite and hematite are present at the four sites. At site 7, the magnetite resides in small round grains and in large skeletal relict crystals. The hematite is fine-grained and associated with biotite, both filling the fractures. Three sites (11–13) were sampled from the intrusive units of the Mascarene Group (Fig. 2); sites 11 and 13 are andesites and site 12 a rhyolite. The andesite is dark, fine-grained, and fresh. The rhyolite is purple to brown and shows no fractures. Cumming's (1967) map indicates that the rock unit sampled at site 13 is probably slightly older than those sampled at sites 11 and 12.

## Methodology

Four to eight cores (2.5 cm diameter) per site were drilled in the field to yield a total of 79 cores from those 13 sites, with one to three specimens (height = 2.2 cm) cut from each core. Experiments were carried out on 145 specimens in quasi-zero residual fields (<2 nT) (Roy *et al.* 1969). Three cleaning techniques were used: thermal, alternating field (AF), and two-stage, the latter consisting of AF followed by thermal. All specimens were subjected to detailed experimental work. For the AF experiments, incremental steps of 5 mT were used up to 50 mT, steps of 10 mT from 50 to 100 mT, and thereafter steps of 25 mT up to 275 mT; a 290 mT and additional intermediate steps were also often used. For the thermal experiments, incremental steps of 100°C were used up to 500°C and thereafter 550, 600, 625, 650, 670, 680, and 690°C. In two-stage work, the transition from AF to thermal was carried out between the 10 and 200 mT stages depending on previous results. After transition, the schedule of incremental thermal steps described above was normally followed. Treatment was discontinued when it became obvious after three or four treatments that no significant amount of remanence of any stability remained in the specimen.

## Results

Although the collection is composed of a wide variety of rock types, some general comments can be made about the remanences found in these rocks.

A steeply inclined remanence ($R_{SP}$) with very high unblocking temperatures ($T_{UB}$)[2] and medium to high coercive forces (rcf) is present at all 13 sites. At eight sites (1–5, 8, 9, and 12) both polarities of the magnetic vector are observed. The reverse polarity only could be detected at site 13. At the remaining four sites (6, 7, 10, and 11) the $R_{SP}$ with normal polarity could be clearly isolated. However, although it could not be established with certainty for reasons given later, the existence of a reverse $R_{SP}$ in these last four sites cannot be ruled out.

A shallow and negatively inclined (≈018°, −27°) remanence ($R_{SW}$) is clearly present at four sites (3, 4, 5, and 6) of the St. George pluton. In many specimens it coexists with the $R_{SP}$. Its $T_{UB}$ and rcf are comparable to those of the $R_{SP}$ but its intensity is larger so detection is relatively easy. The opposite polarity (≈198°, +27°) was also detected in some specimens at each of the four sites. Throughout the investigation, there were indications that the $R_{SW}$ might be present at some other sites.

The complexity resulting from the presence of two remanences with dual polarities is illustrated by the wide scatter of natural remanent magnetization (NRM) directions (Fig. 3). It is noted that there is no grouping about the direction of the present earth's field at the locality (340°E, 73°N). Furthermore, the remanences whose directions happen to lie close to that direction do not behave any differently from other remanences. Recent remagnetization is therefore discounted as a viable explanation for any of these directions.

Because the $R_{SP}$ and $R_{SW}$ have comparable stability spectra, much experimental work was required to separate them. All specimens were not treated at the same time and a particular treatment run usually consisted of two or three specimens from each site. The behaviour of the magnetic vector was understandably quite variable amongst specimens and a description for each specimen would, of course, be too lengthy. As much infor-

[2] $T_{UB}$ is introduced here to describe the temperature at which the remanence is unblocked in laboratory experiments. Although the $T_{UB}$ may correspond to the $T_B$ (blocking temperature) for simple TRM (thermoremanent magnetization) as shown by Thellier (1966), the two can differ greatly if the remanence is of some other origin (e.g., depositional (DRM), chemical (CRM), thermo-chemical (T-CRM)). For example, the temperature at which a remanence of chemical origin unblocks may have very little relationship with the temperature at which the mineral was formed and magnetized since the treatment method differs from the mode of acquisition. Therefore, in order to avoid any implications about the origin of magnetization, which cannot be determined from thermal experiments alone, it appears preferable to distinguish between the experimental observations ($T_{UB}$) and the unknown (or assumed) $T_B$.

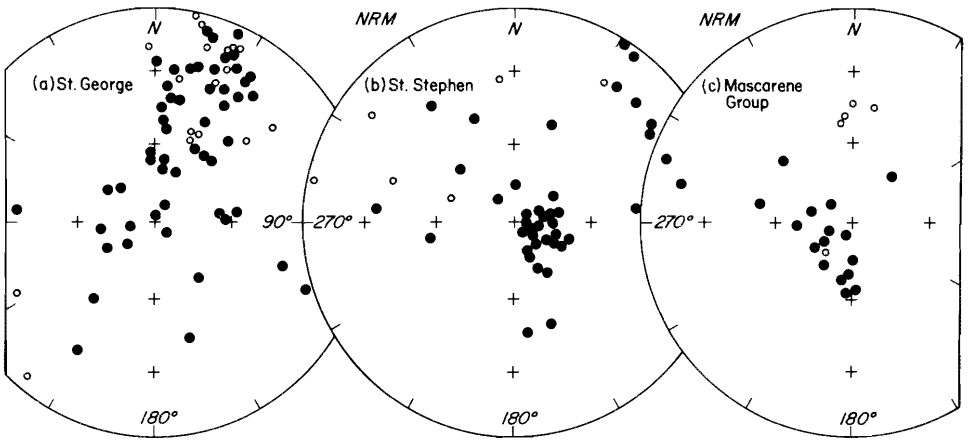



FIG. 3. Specimen directions of NRM (before treatment) of the three rock units. Closed (open) symbols indicate positive (negative) inclinations. Equal area projection.

mation as possible has been compiled in Tables 2 and 3. Because specimens from the same core were normally subjected to different treatments (one to thermal, the other to AF), the specimen has been taken as unit weight. $N_0$ gives the number of specimens that were treated and $N_1$ the number of specimens that were used to determine the $R_{SP}$ and (or) the $R_{SW}$. The number of $R_{SP}$ determinations is given by $N_3$ (Table 2) and the number of $R_{SW}$ by $N_5$ (Table 3). For example, at site 5, $N_0 = 10$; $N_1 = 9$; $N_3 = 6$; and $N_5 = 8$. $N_0 - N_1 = 1$ indicates that, for one specimen, neither $R_{SP}$ nor $R_{SW}$ could be isolated. From the other nine specimens, six $R_{ST}$ and eight $R_{SW}$ determinations could be obtained. The AF, thermal, and AF followed by thermal results have been listed separately together with the $T_{UB}$ and rcf. A single number in the $T_{UB}$ and rcf column indicates that the mean direction was obtained by computing the results after that particular treatment (option (*a*) discussed later in the section 'Direction Determinations'). In most instances, the $T_{UB}$ and rcf are not the same for all specimens and often occupy a certain range. It will be shown with vector diagrams that it is usually difficult to define *precisely* a direction from a single observation. The mean direction was then computed from the treatment range whose limits are given in the treatment column following the option (*c*) procedure described in the section 'Direction Determinations'. For the AF + thermal, the first number indicates that one of the specimens from that group was first subjected to AF treatment up to that field in milliteslas; the second number indicates the maximum $T_{UB}$ subsequently found for that remanence. This treatment was particularly efficient for recovering the $R_{SP}$ from under the larger intensity $R_{SW}$, as will be shown with a few samples later. The $T_{UB}$ is shown with a superscript degree sign so that it can be easily distinguished from the rcf.

### *St. George Pluton*

#### *Sites 3–6*

The most complex behaviour is found at these sites where both $R_{SP}$ and $R_{SW}$ are frequently detected in the same specimen. Three specimens were abandoned because of extremely complex behaviour. In most of the remaining 49 specimens, the two remanences are present and can be clearly defined (Figs. 4, 5), even if the $R_{SP}$ is much weaker than the $R_{SW}$ (compare Int. in Tables 2 and 3). In only two specimens (one from site 5 and one from site 6) does the $R_{SW}$ appear to be missing (Fig. 5*a*,*b*). In both instances the NRM's were weak and below the usual level of intensity of $R_{SW}$ (compare the intensity in Fig. 5*b* with those of Table 3). The $R_{SP}$ was clearly recognized in 34 specimens (Figs. 6, 7).

$R_{SP}$ may also be present in some of the remaining 15 specimens, but its presence could not be established with certainty for a number of reasons. In some instances a steeply inclined remanence was obtained after a treatment, but upon subsequent higher treatments no stable remanence could be observed. It is therefore possible that the $T_{UB}$ or rcf spectrum of that remanence was so restricted that the increment steps used were too large to ascertain its presence from successive treatments. This occurred mostly at the beginning of the study when single-stage (thermal or AF) treatments were used. The two-stage treatment appears to be much more efficient; an explanation by modelling was offered in a previous study yielding similar results (Roy and Lapointe 1976). According to that model, the AF treatment reduces the intensity of the predominant

TABLE 2. Analysis of the steep direction of magnetization $R_{SP}$ determined by AF, thermal, and two-stage (AF followed by thermal) treatments

| | | NRM | | | | Treated | | | | | | | | | | Pole |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Site | $N_0$ | $N_1$ | Int. (mA/m) | $D°, I°$ | κ | Treatment | $N_2$ | Int. (mA/m) | $D°, I°$ | PO | κ | $T_{UB}$; rcf | $N_3$ | $D°, I°$ | κ | $\alpha_{95}$ | Long., Lat. |
| | | | | | | St. George pluton (A) | | | | | | | | | | | |
| 1 | 9 | 9 | 2.3 | 167, +86 | 9 | AF | 3 | 1.3 | 166, +80 | N | 18 | 50–125 | | | | | |
| | | | | | | Th | 4 | 4.7 | 243, +83 | M | 45 | 400°–550° | | | | | |
| | | | | | | AF + th | 2 | 3.5 | 178, +82 | N | — | 15; 550° | 9 | 197, +83 | 38 | 8 | 108°E, 33°S |
| 2 | 12 | 12 | 200 | 032, +56 | 4 | AF | 5 | 33.2 | 078, −68 | R | 49 | 100–240 | | | | | |
| | | | | | | Th | 1 | 110.0 | 079, −54 | R | — | 500° | | | | | |
| | | | | | | AF + th | 3 | 43 | 074, −61 | R | 343 | 10; 580° | 9 | 076, −64 | 70 | 6 | |
| | | | | | | Th | 3 | 34 | 255, +61 | N | 19 | 560°–610° | | | | | |
| | | | | | | AF + th | 6 | 17 | 292, +72 | N | 13 | 240; 640° | 9 | 275, +69 | 15 | 14 | |
| | | | | | | | | | | | | | 18 | 265, +67 | 24 | 7 | 064°E, 30°S |
| 3 | 14 | 13 | 115 | 029, +39 | 8 | Th | 4 | 14 | 295, +76 | M | 8 | 590°–690° | | | | | |
| | | | | | | AF + th | 7 | 9 | 268, +71 | M | 20 | 200; 590° | 11 | 276, +73 | 14 | 12 | 071°E, 40°S |
| 4 | 14 | 13 | 53 | 026, +17 | 11 | AF | 3 | 4 | 079, −77 | R | 17 | 50–200 | | | | | |
| | | | | | | Th | 4 | 2 | 090, +85 | M | 17 | 550°–640° | | | | | |
| | | | | | | AF + th | 1 | 6 | 099, +60 | N | — | 75; 590° | 8 | 043, +89 | 12 | 17 | 115°E, 47°S |
| 5 | 10 | 9 | 957 | 008, +20 | 2 | AF | 1 | 9 | 034, −80 | R | — | 500 | | | | | |
| | | | | | | Th | 2 | 7 | 258, +80 | N | — | 550°–580° | | | | | |
| | | | | | | AF + th | 3 | 10 | 244, +77 | N | 14 | 250; 590° | 6 | 245, +80 | 23 | 14 | 091°E, 35°S |
| 6 | 14 | 14 | 567 | 010, −04 | 3 | AF | 1 | 8 | 271, +69 | N | — | 225 | | | | | |
| | | | | | | Th | 4 | 23 | 278, +69 | N | 27 | 580–640° | | | | | |
| | | | | | | AF + th | 4 | 31 | 260, +76 | N | 46 | 75; 660° | 9 | 270, +72 | 40 | 8 | 071°E, 37°S |
| | | | | | | St. Stephen pluton (B) | | | | | | | | | | | |
| 7 | 17 | 17 | 1721 | 098, +79 | 175 | AF | 7 | 330 | 079, +74 | N | 70 | 20 | | | | | |
| | | | | | | Th | 7 | 321 | 104, +75 | N | 38 | 580°–680° | | | | | |
| | | | | | | AF + th | 3 | 137 | 115, +73 | N | 302 | 20; 680° | 17 | 096, +74 | 54 | 5 | 149°E, 36°S |
| 8 | 11 | 11 | 247 | 041, +18 | 3 | AF | 6 | 12 | 058, +72 | M | 24 | 20–150 | | | | | |
| | | | | | | Th | 4 | 43 | 074, +54 | N | 6 | 500° | | | | | |
| | | | | | | AF + th | 1 | 7 | 108, +60 | N | — | 10; 570° | 11 | 071, +66 | 11 | 14 | 172°E, 43°S |
| 9 | 7 | 7 | 234 | 293, +57 | 3 | AF | 4 | 6 | 282, +76 | M | 14 | 40–100 | | | | | |
| | | | | | | Th | 2 | 2 | 238, +77 | N | — | 550°–560° | | | | | |
| | | | | | | AF + th | 1 | 2 | 068, −79 | R | — | 10; 570° | 7 | 265, +78 | 24 | 12 | 082°E, 39°S |
| 10 | 10 | 10 | 818 | 127, +79 | 18 | AF | 3 | 17 | 132, +73 | N | 20 | 10–90 | | | | | |
| | | | | | | Th | 4 | 228 | 134, +82 | N | 69 | 550° | | | | | |
| | | | | | | AF + th | 3 | 69 | 020, +84 | N | 293 | 20; 600° | 10 | 121, +79 | 37 | 8 | 129°E, 36°S |
| | | | | | | Mascarene Group intrusives (C) | | | | | | | | | | | |
| 11 | 9 | 7 | 1312 | 208, +82 | 21 | AF | 3 | 110 | 154, +69 | N | 38 | 20 | | | | | |
| | | | | | | Th | 4 | 586 | 164, +65 | N | 82 | 550°–600° | 7 | 160, +67 | 64 | 8 | 126°E, 06°S |
| 12 | 9 | 7 | 244 | 235, +81 | 25 | AF | 4 | 13 | 177, +72 | M | 10 | 125–290 | | | | | |
| | | | | | | Th | 3 | 17 | 220, +58 | N | 37 | 550°–600° | 7 | 202, +67 | 13 | 18 | 099°E, 07°S |
| 13 | 9 | 9 | 462 | 233, +61 | 4 | AF | 5 | 58 | 326, −54 | R | 116 | 40–75 | | | | | |
| | | | | | | Th | 2 | 108 | 342, −50 | R | — | 590° | | | | | |
| | | | | | | AF + th | 2 | 280 | 322, −55 | R | — | 20; 560° | 9 | 329, −54 | 52 | 7 | 138°E, 06°N |

NOTES: The site-mean directions prior to treatment (NRM) are also given. $N_0$ = number of treated specimens; $N_1$ = number of specimens used for $R_{SP}$ (this table) and $R_{SW}$ (Table 3) determination. $N_2$ is the number of independent AF, thermal, or AF + thermal observations, and $N_3$ is the total of $N_2$ per site. $D°$ and $I°$ are the east declination and inclination respectively. Int. is the mean intensity. κ = precision estimate and $\alpha_{95}$ = radius of circle of confidence at $P = 0.05$. $T_{UB}$ and rcf indicate the thermal and AF treatment (respectively) after which the mean directions were computed. PO indicates the polarity obtained from a set of observations: N = normal or positive, R = reverse or negative inclination, M = mixed or both polarities were present. For site 2, the normal and reverse polarity results have been compiled separately and then combined. The pole derived from each site mean is given.

TABLE 3. Analysis of the shallow direction of magnetization ($R_{SW}$) determined by AF, thermal, and two-stage treatments

| Site | Treatment | $N_4$ | Int. (mA/m) | $D°, I°$ | κ | $T_{UB}$; rcf | $N_5$ | $D°, I°$ | κ | $\alpha_{95}$ | Pole Long., Lat. |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | St. George pluton (D) | | | | | | | |
| 2* | AF + th | 1 | 28 | 022, −25 | | 10; 580° | 1 | 022, −25 | | | |
| 3 | AF | 6 | 13 | 018, −26 | 13 | 50–100 | | | | | |
| | Th | 7 | 37 | 010, −25 | 15 | 550°–640° | 13 | 014, −25 | 15 | 11 | 097°E, 30°N |
| 4 | AF | 5 | 7 | 022, −16 | 10 | 50–75 | | | | | |
| | Th | 6 | 19 | 021, −14 | 17 | 550°–610° | | | | | |
| | AF + th | 2 | 9 | 022, −25 | — | 15; 560° | 13 | 024, −16 | 14 | 12 | 085°E, 32°N |
| 5 | AF | 3 | 19 | 020, −19 | 13 | 10–150 | | | | | |
| | Th | 3 | 84 | 006, −16 | 9 | 600°–630° | | | | | |
| | AF + th | 2 | 58 | 014, −34 | — | 50; 690° | 8 | 014, −22 | 14 | 15 | 097°E, 32°N |
| 6 | AF | 6 | 104 | 009, −43 | 150 | 100-150 | | | | | |
| | Th | 5 | 565 | 021, −45 | | 550° | | | | | |
| | AF + th | 2 | 172 | 017, −41 | | 75; 600° | 13 | 018, −43 | 42 | 6 | 096°E, 18°N |

*Not used for composite mean (D) in Table 4 because $N_5 = 1$.
NOTES: The site-mean directions prior to treatment are given in Table 2. Headings as defined in Table 2. Definition of $N_4$ and $N_5$ corresponds to Table 2 definition of $N_2$ and $N_3$ respectively.

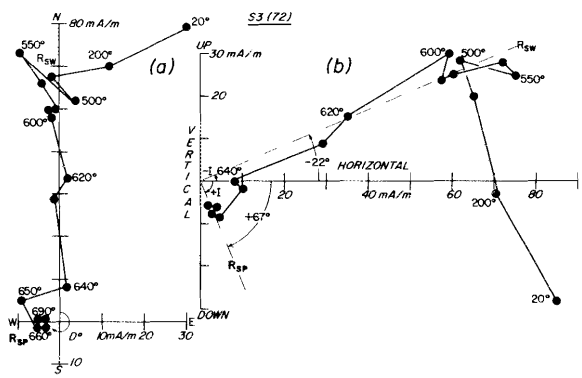



FIG. 4. Vector diagrams showing changes in both direction and intensity after treatment at successively higher temperatures in residual fields <2 nT. The dots represent the termination of each vector. The horizontal component of the magnetic vector is shown in (*a*); the declination *D* is given by the angular distance from the positive *x* (or *N*) axis and the intensity is given by the distance of the dot from the origin. The magnetic vector of the total intensity is represented in (*b*) as a function of the horizontal and vertical components; the intensity is given by the distance of the dot from the origin and the inclination ($\pm I$) by the angular distance from the horizontal axis. The presence of the two remanences ($R_{SW} \approx 000°, -22°$ and $R_{SP} \approx 270°, +67°$) and their relative intensities are clearly shown. In this and following figures, the number following 'S' identifies the site and the number in parentheses, the specimen.

remanence ($R_{SW}$) and consequently lowers the $R_{SW}/R_{SP}$ intensity ratio prior to the thermal treatment; the effect being to optimize the availability of the $R_{SP}$ by widening the $T_{UB}$ window through which the $R_{SP}$ can be recognized. Another difficulty arises from the fact that a $R_{SW}$ of normal polarity (≈220°, +40°) is present in some specimens (e.g., specimen 85B, Fig. 8). In a few instances this creates some uncertainty as to whether the remanence is normal $R_{SW}$ or a normal $R_{SP}$ (Fig. 9). It may well be that the two remanences are still present and this could account for the direction changes observed from 250 mT ($R_{SW}$?) to 590°C ($R_{SP}$?) (Fig. 9). Likewise, the vector changes observed in 90B (Fig. 7) from 560–600°C might be produced by the remnant of a negative $R_{SW}$ as shown by vector subtraction of the 570 and 580°C vectors. Finally, since the $R_{SP}$ usually accounts for only a few percent of the NRM's (0.5% in specimen 84B, Fig. 9), it could remain undetected. It is therefore quite possible that the $R_{SP}$ is present in more than the 34 of the 49 specimens. Conversely, in some specimens, no evidence whatsoever could be found for a $R_{SP}$: for example, in specimen 82A (Fig. 5*c*,*d*) even a two-stage treatment up to 680°C failed to reveal a $R_{SP}$. It is interesting to note that the $R_{SP}$ was clearly identified in its companion specimen 82B (Fig. 5*e*,*f*); this best illustrates the complex magnetic behaviour of the specimens from these sites. The frequency of occurrence of the $R_{SP}$ at these four sites is therefore between 69 and 100%.

*Sites 1 and 2*

The $R_{SP}$ was found in all 21 specimens from these two sites (Fig. 10). The $R_{SW}$ was identified in only one specimen, as shown in Fig. 10*c*, *d*. Of much interest is the fact that at site 2, six of the 12 specimens yielded both polarities while three normal and three reverse polarities were detected in the remaining six specimens (Fig. 10*b*). The means for the reverse and normal polarity directions have been calculated both separately and together (Table 2). Figure 10 also illustrates another difficulty inherent to any paleomagnetic study. What rejection criteria should be used? In this

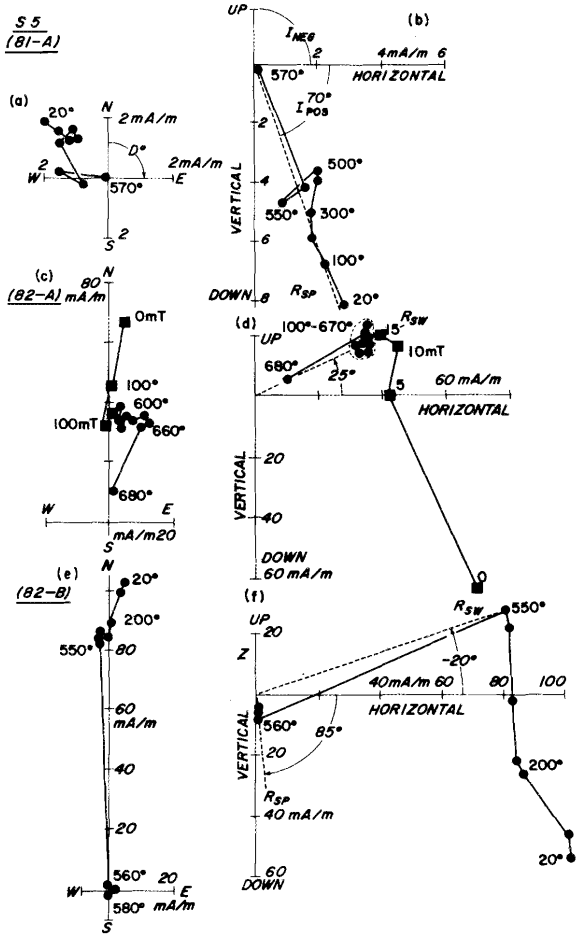



FIG. 5. Vector diagrams of three specimens from site 5 showing the different behaviours encountered within a site. In specimen 81A, only the steeply inclined remanence ($R_{SP}$) is apparently present, (*a*) and (*b*); in specimen 82A, only the shallowly inclined remanence ($R_{SW}$) could be detected, (*c*) and (*d*); in 82B, both remanences are clearly present, (*e*) and (*f*). In this and subsequent figures up to Fig. 12, a thermal treatment is symbolized by a dot (or circle) and an AF treatment by a square.

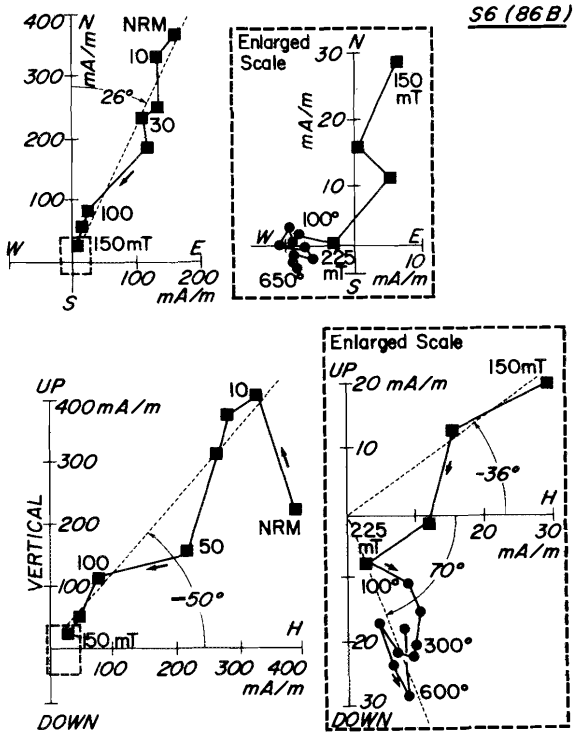



FIG. 6. Vector diagrams showing the large intensity ratio of $R_{SW}/R_{SP}$. The behaviour of the small underlying $R_{SP}$ is shown in detail with the enlarged scale. The two-stage treatment shows that this $R_{SP}$ has rcf > 225 mT and $T_{UB}$ > 650°C. It is noted that the inclination of the $R_{SW}$ at this site 6 is markedly steeper than that found at the other sites (Table 3; Fig. 5).

study the rejection was kept to a minimum. Hence, the two directions 002°, +70° and 350°, +53° in Fig. 10*b*, which are significantly different from the other seven reasonably well clustered directions, were not rejected for the computation of the results (Table 2). If those two directions had been rejected, the positive polarity and the combined results would have been $N = 7$; 258°, +64°; $\kappa = 58$; $\alpha_{95} = 8°$ and $N = 16$; 256°, +64°; $\kappa = 68$; $\alpha_{95} = 5°$ respectively. Comparison with the tabled results for site 2 shows that the mean direction would have been different by only 3° of arc; the statistics, however, would have been much improved with $\kappa$'s of 58 and 68 instead of 15 and 24 respectively.

In summary, both the $R_{SP}$ and the $R_{SW}$ are frequently found in the St. George pluton, that is, in 79 and 67%, respectively, of the 70 specimens studied.

*St. Stephen Pluton*

The $R_{SP}$ was obtained from all specimens of the four sites (Fig. 11). Both polarities were found at sites 8 and 9 and it is possible that the scatter of NRM's (Fig. 3*b*) is produced by the resultant of dual polarity remanences (Fig. 11*d*,*e*). No evidence could be uncovered for the $R_{SW}$ found in the St. George pluton. At sites 8, 9, and 10 the $T_{UB}$ are usually in the 550–660°C and the rcf in the 90–150 mT range. The peridotite (site 7) has remarkably high $T_{UB}$ (up to 680°C), but its rcf is only 20–30 mT. For this reason, the high $T_{UB}$ were much more easily and accurately defined when the specimens were first subjected to AF treatment of 20 mT (Table 2). It is noteworthy that the intensity remaining at 650°C is extremely high (380 mA/m, Fig. 11*a*). Most of the specimens from these four sites have large NRM intensities (>200 mA/m, Table 2); at 500°C the intensities are usually >100 mA/m (Fig. 11*a*) and the directions steeply inclined (Fig. 11*b*, *c*). In a few specimens the intensities are much weaker but the behaviour of

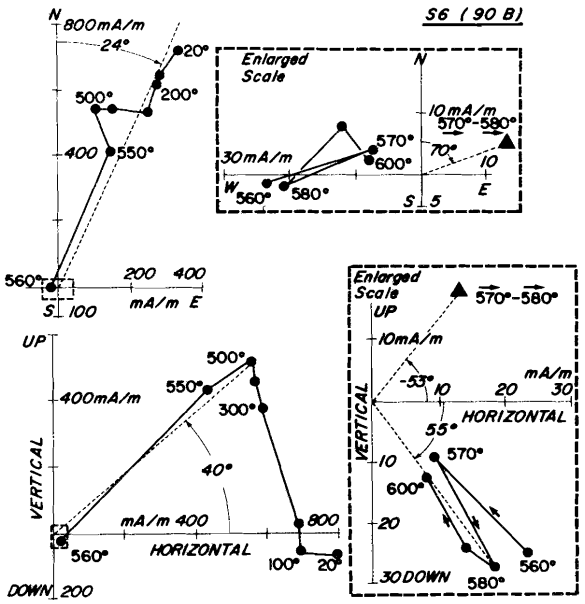



FIG. 7. The details shown in the enlarged scale permit one to detect the dual polarity of the $R_{SP}$. The presence of a reverse $R_{SP}$ is readily observed; the intensity being much smaller after the 570°C treatment than after the 580°C treatment. The result of the vector subtraction (570 − 580°, triangle) shows that the inclinations of the normal and reverse $R_{SP}$ are within 02° of antiparallelism (bottom right).

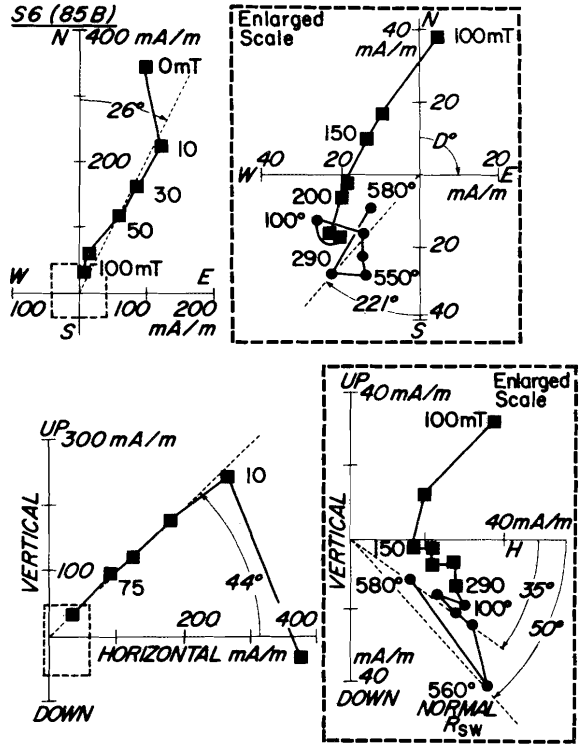



FIG. 8. Vector diagrams showing a $R_{SW}$ with dual polarity. The declinations are about 15° off antiparallelism. The inclination of the normal $R_{SW}$ is difficult to determine accurately but is between 35 and 50°, in good agreement with the −44° inclination of the reverse $R_{SW}$. The potential presence of this normal $R_{SW}$ sometimes creates difficulties in distinguishing between a normal $R_{SW}$ and a normal $R_{SP}$ whose mean direction is 261, +68 (Table 4, St. George pluton (A)). This problem is illustrated in Fig. 9.

direction changes is similar (Figs. 3, 11*d*). In the absence of any other stable remanence this $R_{SP}$ is believed to have been acquired during cooling following emplacement. Because some of the remanence is evidently carried by hematite ($T_{UB} > 675$°C in some specimens, at least), the remanence is probably a T-CRM (thermo-chemical remanent magnetization).

*Mascarene Group Intrusives*

A $R_{SP}$ is also found in 23 of the 27 specimens from these three sites. After treatment, the polarity of the direction is normal at sites 11 and 12 except for one reversely polarized specimen of site 12. All specimens from site 13 are reversely polarized and the mean direction is slightly less steeply inclined (−54°) than at the other two sites (+67°) (Table 2). This could be indicative of a polar shift as discussed later. Most of the scattered NRM directions of this group (Fig. 3*c*) belong to site 13. This scatter might be attributable to the presence of a normal $R_{SP}$ with low $T_{UB}$ and low rcf as shown by vector analysis (Fig. 12*c*, *d*).

The $T_{UB}$ for this group are from 500–600°C so that hematite is the carrier of part of the remanence. This is supported by the fact that the rcf's occasionally are as high as 290 mT (Fig. 12*b*). There are no appreciable direction changes between 550 and 600°C, so that if magnetite is the carrier of the major part of the remanence, as it might be suspected owing to the large intensities (Fig. 12*d*; Table 2), then both magnetite and hematite were formed contemporaneously, most probably at the time of emplacement.

*Direction Determinations*

The details shown in vector diagrams (Figs. 4–12) serve to draw attention to the difficult task of determining the direction of a magnetization and the problems involved in finding the most representative data set. Even when a remanence ($R_{SW}$ or $R_{SP}$ in this study) is identified by a group of vectors, it is evident from the scatter of vector terminations that the direction cannot be defined precisely but can only be *estimated*. The question is, How can it be estimated best? Several options are open: (*a*) only the vector obtained after the last treatment yielding that direction is accepted, (*b*) the

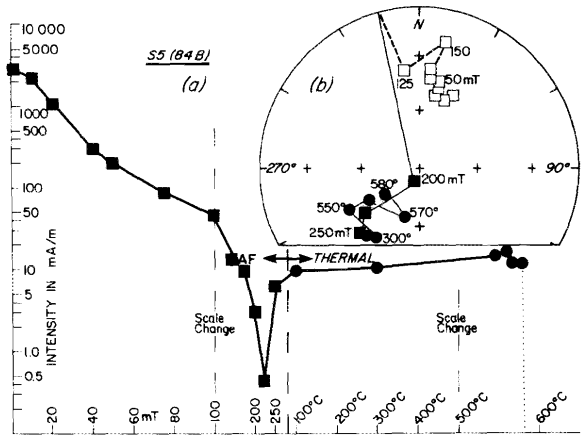



FIG. 9. An example showing one of the difficulties that is sometimes encountered in ascertaining which remanence is present. The direction obtained between 250 mT and 300°C is nearly antiparallel (within 25° of arc) to the reverse $R_{SW}$ ($\leq$125 mT) and it would appear that most of the remaining remanence is attributable to a normal $R_{SW}$. Note that the intensity is on a logarithmic scale and that the low intensities correspond to the crossing over from reverse to normal polarity. However, upon further treatment the direction steepens and moves toward the normal $R_{SP}$ (mean = 245°, +80° for this site). It is therefore possible that both remanences are still present. The dotted line is simply to indicate the following treatment (590°C) after which no stable remanence could be detected.

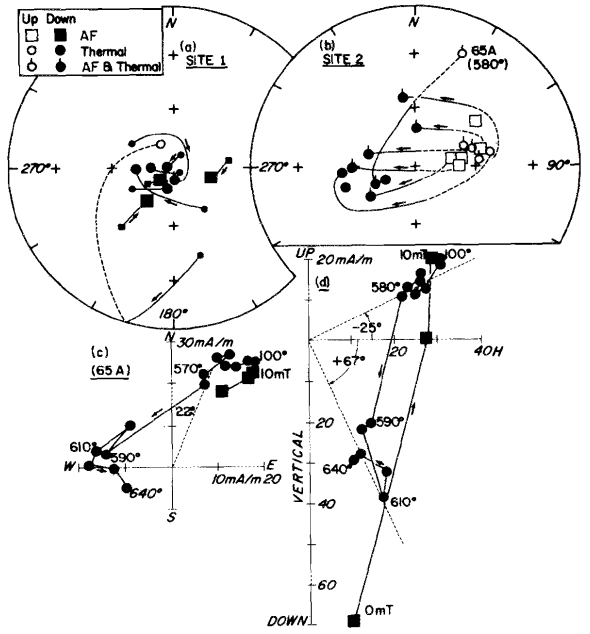



FIG. 10. Directions of remanence at sites 1 and 2. In (*a*) the NRM directions are shown by small symbols and directions after treatment by large symbols. The treatment to which the specimens were subjected is indicated. In (*b*) only the directions obtained after treatment are shown. For six specimens, both polarities were obtained and those are joined by a dashed and then full line. The treatment is identified. In three specimens, for example, the negative polarity remanence was obtained by AF treatment and the positive polarity remanence was obtained by subsequent thermal treatment. Only one $R_{SW}$ was identified from these two sites and it is shown in (*c*) and (*d*).

vectors whose directions are essentially the same are averaged, (*c*) the vector directions which are essentially the same are averaged vectorially, giving all vectors unit weight, or (*d*) only the first vector whose direction approximates the directions obtained subsequently is accepted. It is worth noting that the scatter shown on the diagrams would be less apparent on stereonets where only directions are shown.

Option (*a*) can be used under the probability that the last treatment is the most likely to have removed the effect of all other remanences; however, the intensity is often smaller than in previous determinations so that a smaller signal to noise ratio may introduce appreciable instrumental errors into the determination. Conversely, in (*d*) the signal to noise ratio is usually large, minimizing instrumental errors; however, a considerable part of another remanence might still be present. The averaging performed in both (*b*) and (*c*) tends to reduce both causes of error: the signal to noise ratio and the remnant of another remanence. The former will be more effectively reduced using (*b*) and the latter using (*c*). In this study, we have used (*a*) on the few occasions when very little direction change occurred during the previous treatments — the intensities were then usually large (Tables 2, 3). In most instances we have opted for (*c*) and in Fig. 4, for example, the $R_{SW}$ direction has been obtained by averaging the 560–630°C directions and the $R_{SP}$ by averaging the 660–690°C directions. Comparisons using the (*c*) and (*d*) procedures were made on a few specimens and it was found that when the number of determinations is equal to or exceeds three, the mean directions are within a few degrees, usually within 3°.

## Discussion

A steeply inclined remanence ($R_{SP}$) of high stability against both thermal and AF treatments is evidently present in the three rock units sampled. The question arises, Are the directions obtained from these rock units significantly different? When using site unit weight, the circle of confidence ($\alpha_{95}$) of the St. Stephen overlaps both the St. George and the Mascarene Group circles (Fig. 13*d*). This is not surprising since the circles of confidence are large owing to the small number of sites. Furthermore, it has been difficult to determine accurately the direction of magnetization for a number of reasons. Firstly, at least one field reversal occurred at the

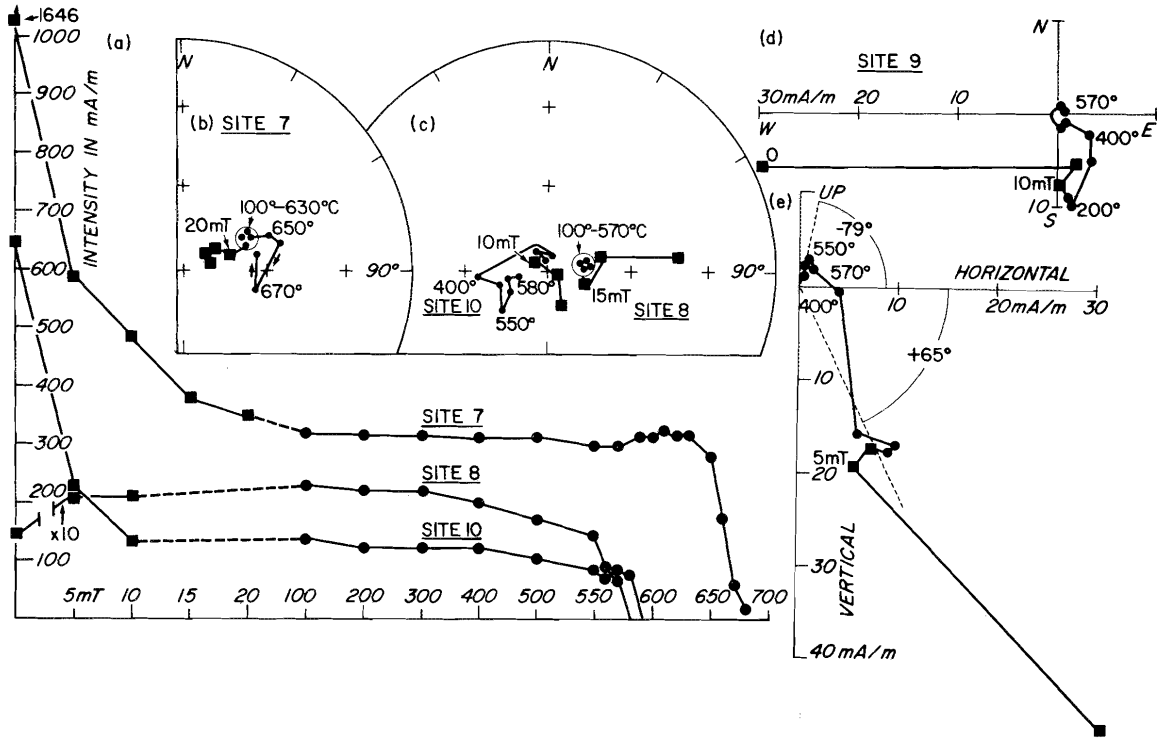



FIG. 11. Examples of two-stage treatment from each of the four St. Stephen pluton sites. Only the $R_{SP}$ could be detected at these sites. The two polarities were clearly present at sites 8 and 9 as shown with an example in (*d*) and (*e*). The intensity of the hematite remanence carried by the peridotite is surprisingly large as shown by the decay curve; site 7 in (*a*).

time of remanence acquisition, as shown by the normal and reverse polarities frequently observed within sites (Table 2) and even within specimens (Figs. 7, 10, 12). The direction of dual polarity remanences can be difficult to define precisely since the pre- and post-reversal magnetizations are likely to have similar characteristics. Consequently, unless the remanence acquired in one polarity clearly predominates over the other in one part of the stability spectrum, the resultant vector may diverge greatly from the direction of the magnetizing field if that field is not exactly antiparallel before and after reversal, and the field is very unlikely to be antiparallel if only because of secular variation (Roy and Park 1974). Secondly, the definition of the $R_{SP}$ in the St. George intrusion is further complicated by the fact that another dual polarity remanence ($R_{SW}$) of high stability (against both thermal and AF treatments) is also present.

All the directions that have been used to calculate the results after treatments are shown in Fig. 13. Most of them form a fairly tight grouping, but some diverge considerably from the mean. These latter might be either the result of unresolved remanences of dual polarity or attributable to incomplete removal of an $R_{SW}$. Whatever the cause, their acceptance certainly contributes to an increase in the radius of the circle of confidence, even though it does not change the rock unit mean appreciably. As a matter of interest, calculations were made using specimen unit weight. The results are as follows:

| | | | |
|---|---|---|---|
| St. George | (A) | Mean = 260°, +75°; | $\alpha_{95}$ = 4.3° |
| St. Stephen | (B) | Mean = 089°, +79°; | $\alpha_{95}$ = 5.4° |
| Mascarene Group | (C) | Mean = 164°, +63°; | $\alpha_{95}$ = 7.2° |
| St. George | (D) | Mean = 016°, −27°; | $\alpha_{95}$ = 5.7° |

Comparison with the Table 4 results shows that the mean directions differ by 3° or less for each rock unit but the $\alpha_{95}$ is reduced drastically (by 50–70%). Those circles of confidence are shown in Fig. 13. *F*-distribution tests also show that the mean directions are significantly different at the 95% confidence limit (Table 4; *F* value > value at $P = 0.05$).

If it is taken that the directions are significantly different, then the pole displacement relative to southwestern New Brunswick was fairly rapid. The Mascarene Group intrusives are Upper Silurian, while both the St. Stephen and the St. George plutons are Lower Devonian (near the Silurian–Devonian boundary); therefore the three rock units were probably emplaced within a few

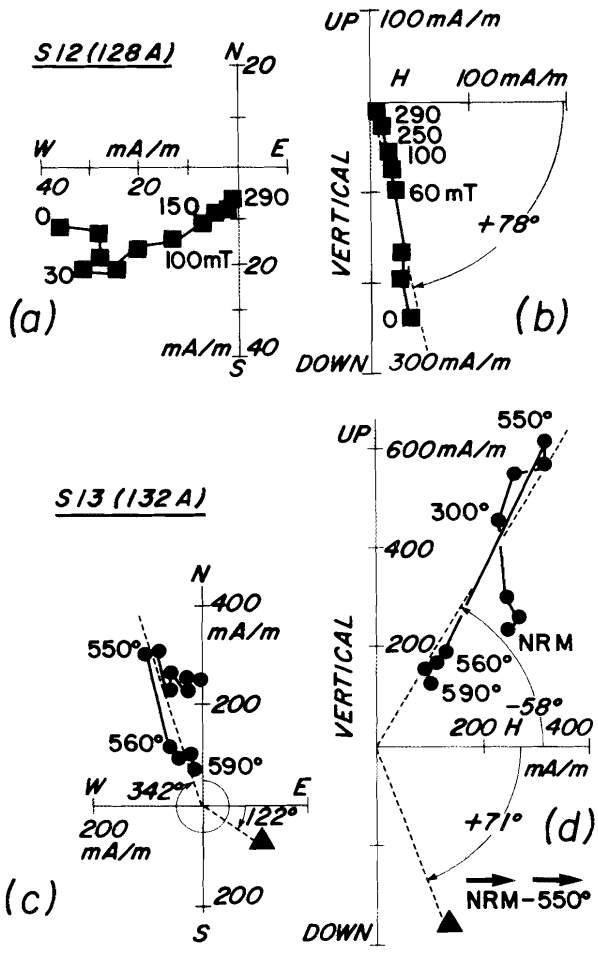



FIG. 12. Examples of vector changes from the Mascarene Group. Little direction changes occur in the normally polarized sites 11 and 12, as shown by an example from site 12 in (*a*) and (*b*), except that the remaining intensity is much larger at site 11 than 12 (Table 2). After treatment, all specimens at site 13 are reversely polarized as shown in (*c*) and (*d*) (see also Fig. 13*c*). Subtraction of the 550°C vector from the NRM vector yields a vector (triangle) nearly antiparallel (within 22° of arc) to the 550°C vector, indicating that this NRM direction (359°, −42°) is the result of a dual polarity remanence, with most of the normally polarized remanence (≈122°, +71°) having $T_{UB}$ between 20 and 550°C, and most of the reversely polarized remanence (≈342°, −58°) having $T_{UB} > 550$°C.

tens of millions of years of one another. Yet, the pole displacement is considerable (Table 4). The interpretation of some of the results obtained here is ambiguous and could possibly be additional evidence for a rapid apparent polar movement. For example, in the Mascarene Group the inclination at site 13, besides being entirely reversed, is 13° shallower than the inclination at sites 11 and 12 (Fig. 13*c*; Table 2). Is this indicative of different ages of intrusion and (or) magnetization? According to Cumming's (1967, Fig. 2) map, site 13 is from Formation B (pre-Silurian or Silurian), and older than the undifferentiated Silurian site 12 and the Pembroke Formation site 11. Likewise, in the St. George pluton, the inclination of the $R_{SW}$ at site 6 (−43°) is markedly steeper than the mean inclination of sites 2–5 (−22°) which show good between-site agreement (Table 3). This is also illustrated by comparing the inclination of the $R_{SW}$ at site 5 (Fig. 5, −25 to −20°) and at site 6 (Fig. 6, −50°; Fig. 7, −40°). Are these differences significant? Evidently the data are not numerous enough to answer these questions and more results are needed. At this stage, we would simply like to point out that a rapid apparent polar wander offers an alternative explanation for the (nongrouped) directions which have been attributed to unresolved remanences. A T-CRM acquired slowly during an interval of rapidly changing ambient field direction would produce results similar to those shown in Fig. 13, and those apparently divergent directions might be recording the field direction change.

In this regard, it is interesting to note that the A ($R_{SP}$) and D ($R_{SW}$) directions found in the St. George pluton are 58° apart; yet, it is impossible to establish with certainty which of the two was acquired first. Both have very high $T_{UB}$ and resist very high AF treatment (Tables 2, 3). Part (at least) of each is carried by hematite so that they are either a CRM or a T-CRM. Upon treatment, the magnetic vector usually moves from D to A (never from A to D) so that it can be said that in general the $R_{SP}$ has higher $T_{UB}$ and rcf than the $R_{SW}$. However, the remanence with the highest $T_{UB}$ or rcf is not necessarily the oldest (Buchan and Dunlop 1976). Furthermore, in some specimens only the $R_{SP}$ is present, while other specimens appear to carry a $R_{SW}$ exclusively (Fig. 5). The contribution of magnetite as a carrier is difficult to establish because some of the remanence with $T_{UB}$ in the 550–570°C range could possibly be carried by the hematite in the ilmo-hematite compositions. If magnetite is a carrier, the larger intensities of the $R_{SW}$ suggest that most of it was magnetized when the field was in the D direction. However, although the intensities of the $R_{SP}$ are usually weak, large intensities are occasionally found so that some of the magnetite might also have been magnetized in the A direction. We are then faced with the problem of two remanences each of which might be carried by both magnetite and hematite. Were there two episodes of magnetization, and if so what mechanism could produce two such remanences? According to Pajari *et al.* (1974), the St. George pluton has not been subjected to a serious thermal event after the initial cooling. So, were the remanences acquired during initial cool-

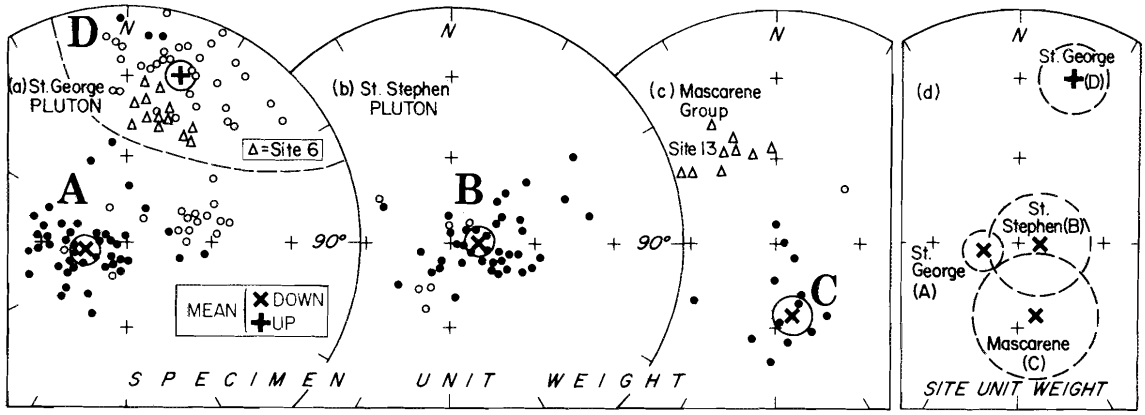



FIG. 13. Synopsis of the directions obtained after treatment. All individual directions used to obtain the four mean directions (A, B, C, and D) are shown in (*a*), (*b*), and (*c*). The polarity of the directions with a negative inclination has been reversed to calculate the means of A, B, and C. The means and (distinctly different) circles of confidence ($\alpha_{95}$) using specimen unit weight are shown in (*a*), (*b*), and (*c*) and given in the text. The mean and (overlapping) circles of confidence ($\alpha_{95}$) using site unit weight are shown in (*d*) and given in Table 4. *F*-distribution tests show that the means are significantly different at $P = 0.05$ (Table 4), indicating that the significance of circles of confidence should be interpreted with reservation.

TABLE 4. Mean directions and poles

| | | | | | | | | *F*-distribution | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | *N* | $D°, I°$ | *R* | κ | $\alpha_{95}$ | Long., Lat. | $\delta p°, \delta m°$ | Pop. | *F** | 0.05† |
| St. George pluton (A) | 6 | 261, +78 | 5.93 | 66 | 8 | 085°E, 38°S | 14, 15 | A, B | 7.87 | 3.63 |
| St. Stephen pluton (B) | 4 | 089, +82 | 3.88 | 26 | 18 | 136°E, 43°S | 35, 35 | A, C | 12.37 | 3.74 |
| Mascarene Group (C) | 3 | 167, +64 | 2.94 | 32 | 22 | 122°E, 02°S | 28, 35 | B, C | 4.51 | 4.10 |
| St. George pluton (D) | 4 | 018, −27 | 3.93 | 43 | 14 | 093°E, 29°N | 09, 15 | C, D | 17.41 | 4.10 |
| A + B | 10 | 241, +87 | 9.62 | 24 | 10 | 106°E, 42°S | 20, 20 | | | |
| A + C | 9 | 212, +79 | 8.62 | 21 | 12 | 100°E, 26°S | 22, 23 | | | |
| B + C | 7 | 145, +77 | 6.66 | 18 | 15 | 128°E, 24°S | 27, 28 | | | |
| A + B + C | 13 | 192, +84 | 12.38 | 19 | 10 | 110°E, 32°S | 20, 20 | | | |

*$F = (N - 2)[(R_1 + R_2 - R)/(N - R_1 - R_2)]$, where $R_1$ and $R_2$ are the resultant vectors of each population and $R$ is the resultant vector of the two populations.
†Value at $P(F) = 0.05$.
NOTE: Composite means are also given although the *F*-distribution test (Watson 1956) indicates that the directions are significantly different at the 95% confidence limit. δp and δm are the semi-axes of the oval of confidence at $P = 0.05$.

ing? This would require an extremely rapid field direction change. Perhaps one of the remanences was acquired during regional hydrothermal alteration (Ade-Hall *et al.* 1971). Whatever the cause, it is clear that the age of one of these remanences is significantly different from the age of intrusion, and although that remanence may be useful to delimit the polar path pattern, it cannot be used for calibration.

## The Lower Paleozoic Poles

All poles obtained from Lower Paleozoic rock units are shown in Fig. 14. To the poles shown in Fig. 1, other poles obtained from this and other recent studies have been added and are identified by a number, and listed in Addendum B. Unnumbered poles can be identified from Fig. 1 and Addendum A. The new poles certainly add a new dimension to the Lower Paleozoic picture. The impression of a stationary field during the Ordovician–Devonian interval depicted by the reasonably tight grouping of those poles at about 120°E, 35°N (Fig. 1) appears badly shattered. Although three of the new poles (54, 55, and 81) fall north of the equator and reasonably close to that grouping, the others are a considerable distance to the south. Before discussing the significance of these new results in more detail, two comments should be made. (1) As for Fig. 1, the poles are symbolized according to the best estimate of the rock age which is not necessarily the age of magnetization. (2) The exact locations of poles lying near Australia and obtained from the eastern half of North America are not easily defined because of the high paleolatitude of the sampling site. In such instances, the semi-axes (δp, δm) of the oval of confidence of the pole are almost twice as large as the radius of confidence ($\alpha_{95}$) of the mean direction

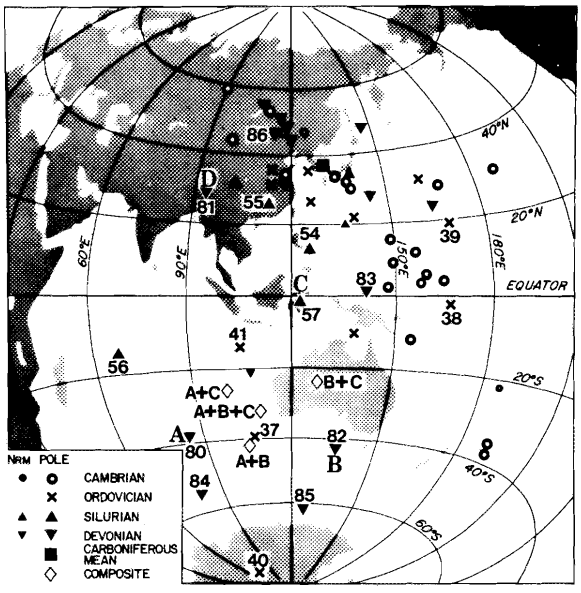



FIG. 14. All Lower Paleozoic poles. Unnumbered poles can be identified from Fig. 1 and Addendum A. The new poles are identified by a number and listed in Addendum B. The poles are symbolized according to the best estimate of the age of the rock unit, which does not necessarily correspond to the age of remanence acquisition. Poles obtained by combining results from the rock units of this study are shown by diamond symbols and identified as per Table 4.

(Table 4). The problem in latitudinal positioning is readily seen from the relation tangent of latitude equals one half tangent of inclination. With a steep magnetic vector ($I$ near 90°) the orientation of the horizontal component ($D$) is difficult to define. Although definition of $D$ affects to a certain extent the latitudinal positioning, it is more serious for longitudinal positioning. Furthermore, because the present grid system is used to calculate poles, the longitudinal problem appears (in terms of degrees) enhanced. For example, although the directions of sites 8 and 9 differ by only 36°, the longitudinal separation of the poles is 90°. The angular separation on a sphere, however, is less than 70° since the latitude is about 40°. A quick inspection of Table 2 shows that the dispersion of site mean poles of the St. George and St. Stephen plutons is much larger longitudinally than latitudinally. Mean poles obtained by combining the results of two or three rock units are given in Table 4 and shown in Fig. 14. It is apparent that whether the results are taken singly or combined, the pole is definitely south of the equator.

Two recent studies of Lower Paleozoic rock units have also yielded multipoles, some of which lie in the southern hemisphere. Murthy and Rao (in preparation) have found diabase dikes from Bonavista Bay (Newfoundland) carrying two distinct remanences yielding poles at about 141°E, 02°N (pole 83) and 080°E, 54°S (pole 84), roughly in agreement with the C (57) and A (80) poles respectively. The exact ages of poles 83 and 84 are uncertain but are believed to be near the Silurian–Devonian boundary (Murthy and Rao, personal communication, 1978). Four poles were obtained by Lapointe (1979) from the Botwood Group and the Mount Peyton Batholith (Newfoundland). The Silurian Botwood Group poles (54 and 55) fall in the general area of other Silurian poles. The Mount Peyton Batholith, which intrudes the Botwood Group, is believed to be a two-phase intrusion with dates of 420 and 380 Ma; the two phases yield distinct poles, 56 and 85, respectively, the latter being in rough agreement with the B(82) pole. We then have four poles obtained from three different studies which lie between 38 and 63°S, some 70–100° south of the main Ordovician–Devonian grouping (Figs. 1, 14). The best estimate of the mean age for these poles is Early Devonian, and it appears that a rapid and huge apparent polar wander took place at that time. Is this polar path applicable to the entire North American craton or to the eastern seaboard only (area A)? The latter must be seriously considered, since the four poles were obtained from Newfoundland and New Brunswick and their southern location might be indicative of relative motion between area A and the North American craton, as suggested by Kent and Opdyke (1978). This alternative will probably remain viable unless similar poles can be found from the North American craton. Conversely, although they cannot be considered as hard evidence, there are some results which make this alternative less attractive. The St. George pluton yields a pole (D or 81) which is close to the Ordovician–Devonian grouping. Therefore, if area A and the North American craton were once separate entities, then it must be assumed that the $R_{SW}$ yielding pole D was acquired after the joining and postdates $R_{SP}$ (pole A). This assumption cannot be confirmed or negated since it could not be established which of the two remanences was acquired first. Of much interest is the remanence carried by the Ordovician Trenton limestone from New York State (McElhinny and Opdyke 1973). The pole (36) obtained after treatment is in good agreement with the other Ordovician poles. However, the pole (37) derived from the NRM is located at 107°E, 40°S (McElhinny and Opdyke 1973), that is, between poles A and B (Fig. 14). Since the NRM is the resultant of the Ordovician remanence and another remanence removed by cleaning, pole 37 is not an accurate, but only a rough, indicator of the latter

remanence. If the effect of the Ordovician remanence is removed, the pole then shifts towards the south. However, since the ratio of intensities, NRM/'after treatment', is very large ($\approx 10$), the contribution of the Ordovician remanence to the NRM is small and its effect on NRM pole location is moderate as shown by vectorially erasing the 'after treatment' result from the NRM result. Vector subtraction using an intensity ratio of 10 shows that the pole moves 11° further to the south (106°E, 51°S). It then appears that the remanence removed by cleaning is also an ancient magnetization which was acquired when the apparent pole, although poorly defined because of the underlying Ordovician remanence, was located in intermediate latitudes. Pole 37 has remained unexplained because it could not be fitted to any known part of the Phanerozoic apparent polar path. We suggest that it may belong to the grouping of poles 80, 82, 84, and 85. If so, it would not favour the separation alternative (or area A must be enlarged to include at least part of New York State). Two remanence directions (groups 1 and 2; Addendum B) were obtained from Ordovician rock units of the Eastern Townships (Province of Quebec) (Séguin 1977). The age of magnetization could not be determined and the poles that have been recalculated with and without correction for tilt are given for both groups in Addendum B and shown in Fig. 14 (poles 38–41). It is interesting to note that the group 2 pole (obtained from well within the North American craton) lies south of the equator, whether it is corrected (pole 41) or not (pole 40) for tilt. It would then appear that the possibility that all of North America was subjected to rapid apparent polar wander cannot be rejected.

## Conclusions

This study shows that although Paleozoic rocks are younger than Precambrian rocks their magnetic history may be as complicated, as shown by the finding of two remanences, both with dual polarity, in the same core. Much experimental work was needed to separate them and vector diagrams were found to be most useful for the interpretation of data; much information initially unnoticed on stereonets was readily seen on this type of display. The results indicate that a rapid apparent polar movement occurred in Siluro-Devonian times. This is supported by two other studies from Newfoundland. Whether this indicates that the entire North American craton was in movement at that time or that the eastern margin was detached from, and in relative movement with, the main craton has yet to be determined.

All the presently available Lower Paleozoic data are shown in Figs. 1 and 14 and listed in Addenda A and B. These have been produced as base maps. We have refrained from drawing an apparent polar path for the following reasons. Firstly, the poles cover too wide an area and are insufficient in number to do so confidently; and secondly, the poles have been plotted according to the best estimate of the age of the rock unit which may not be the age of remanence acquisition so that the age assignment of some poles may be incorrect.

The results obtained in this study do not simplify, but rather show the complexity of, the Lower Paleozoic picture. Nonetheless, they provide one with partial answers to some of the earlier questions. The apparent pole did not remain stationary from Ordovician to Permian, at least for some parts of the present North American continent. If such a large apparent polar wander occurred during Siluro-Devonian times, it is possible that it might have also taken place during the Cambrian, and the stringing of Cambrian poles may not be attributable to remagnetization but to apparent polar wander.

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**Addendum A**

| Key | Rock unit | Age | Pole | Reference |
|---|---|---|---|---|
| 1 | Lamotte Fm. (group IIa) | Camb. | 192°E, 38°S | Al-Khafaji and Vincenz (1971) |
| 2 | St. Joe | Camb. | 190°E, 35°S | Beales *et al.* (1974) |
| 3 | Tapeats sandstone | Camb. | 158°E, 05°N | Elston and Bressler (1977) |
| 4 | Lodore Fm. | Camb. | 186°E, 23°S | Collinson and Runcorn (1960) |
| 5 | Lamotte Fm. (group IIb) | Camb. | 155°E, 12°S | Al-Khafaji and Vincenz (1971) |
| 6 | Wichita granites | Camb. | 164°E, 04°N | Spall (1970) |
| 7 | Lamotte Fm. (group I) | Camb. | 187°E, 30°N | Al-Khafaji and Vincenz (1971) |
| 8 | Wilberns Fm. | Camb. | 159°E, 06°N | Van der Voo *et al.* (1976) |
| 9 | Colorado intrusives | Camb. | 156°E, 12°N | French *et al.* (1977) |
| 10 | Wichita Mountains | Camb. | 147°E, 02°N | Spall (1968) |
| 11 | Bradore Fm. | Camb. | 149°E, 09°N | Black (1964) |
| 12 | Bradore Fm. | Camb. | 167°E, 29°N | Rao and Deutsch (1976) |
| 13 | Wichita Mountains | Camb. | 149°E, 16°N | Vincenz *et al.* (1975) |
| 14 | Wichita Mountains | Camb. | 138°E, 30°N | Vincenz *et al.* (1975) |
| 15 | Lamotte Fm. (locality E) | Camb. | 137°E, 32°N | Al-Khafaji and Vincenz (1971) |
| 16 | Wichita Mountains | Camb. | 134°E, 34°N | Ku *et al.* (1967) |
| 17 | Wichita Mountains | Camb. | 118°E, 34°N | Spall (1968) |
| 18 | Sawatch sandy dolomite | Camb. | 125°E, 47°N | Howell and Martinez (1957) |
| 19 | Deadwood Fm. | Camb. | 117°E, 47°N | Collinson and Runcorn (1960) |
| 20 | Paleozoic plutons,* central Colorado | Camb. | 099°E, 44°N | Larson and Mutschler (1971) |
| 21 | Paleozoic plutons,* central Colorado | Camb. | 122°E, 37°N | Larson and Mutschler (1971) |
| 22 | Abrigo Fm., Arizona | Camb. | 089°E, 59°N | Elston and Bressler (1977) |
| 23 | Muav Limestone, Grand Canyon | Camb. | 110°E, 55°N | Elston and Bressler (1977) |
| 30 | Lower Paleozoic dikes, Grenville Province | Ord. | 138°E, 10°S | Palmer *et al.* (1977) |
| 31 | Queenston Fm. | Ord. | 161°E, 31°N | Peterson (1975) |
| 32 | Steel Mtn., Nfld. | Ord. | 139°E, 22°N | Murthy and Rao (1976) |
| 33 | Newfoundland Zinc Co. mine, Mississippi Valley-type ore deposit | Ord. | 126°E, 26°N | Beales *et al.* (1974) |
| 34 | Beemerville alkaline complex | Ord. | 126°E, 35°N | Proko and Hargraves (1973) |
| 35 | Juniata Fm. | Ord. | 114°E, 32°N | Van der Voo and French (1977) |
| 36 | Trenton limestone | Ord. | 114°E, 36°N | McElhinny and Opdyke (1973) |
| 50 | Rose Hill Fm., Maryland | Sil. | 136°E, 20°N | Graham (1949) |
| 51 | Clinton iron ore, Alabama | Sil. | 139°E, 34°N | Howell *et al.* (1958) |
| 52 | Bloomsburg Fm., Penn. | Sil. | 102°E, 32°N | Roy *et al.* (1967) |
| 71 | Columbus limestone and Raisin River dolomite, Ohio | Dev. | 164°E, 24°N | Martin (1975) |
| 72 | Clam Bank Group | Dev. | 146°E, 28°N | Black (1964) |
| 73 | Onondaga limestone, N.Y. | Dev. | 107°E, 21°S | Graham (1956) |
| 74 | Perry Fm. red beds, N.B. | Dev. | 118°E, 32°N | Robertson *et al.* (1968) |
| 75 | Columbus limestone (exclusive of transition zone) | Dev. | 120°E, 45°N | Martin (1975) |
| 76 | Delaware limestone | Dev. | 118°E, 48°N | Martin (1975) |
| 77 | Belchertown pluton | Dev. | 147°E, 48°N | Ashwall and Hargraves (1977) |
| 78 | Becker Butte, Arizona | Dev. | 109°E, 56°N | Elston and Bressler (1977) |
| 79 | Temple Butte, Grand Canyon | Dev. | 115°E, 53°N | Elston and Bressler (1977) |
| 100 | Mean pole | Carb. | 130°E, 37°N | Roy (1969) |

*May be upper Ordovician.

## Addendum B

| Key | Rock unit | Age | Pole | Reference |
|---|---|---|---|---|
| 37 | Trenton limestone | Ord. | 107°E, 40°S | McElhinny and Opdyke (1973) |
| 38 | Group 1* WRH (14°) | Ord. | 166°E, 02°S | Séguin (1977) |
| 39 | Group 1* WRB (20°) | Ord. | 168°E, 19°N | Séguin (1977) |
| 40 | Group 2* WRH (39°) | Ord. | 070°E, 81°S | Séguin (1977) |
| 41 | Group 2* WRB (38°) | Ord. | 105°E, 14°S | Séguin (1977) |
| 54 | Botwood Group (rhyolite) | Sil. | 125°E, 13°N | Lapointe (1979) |
| 55 | Botwood Group (sandstone) | Sil. | 113°E, 25°N | Lapointe (1979) |
| 56 | Mount Peyton (diorite) | Sil. | 068°E, 15°S | Lapointe (1979) |
| 57 | Mascarene Group (C) | Sil. | 122°E, 02°S | This paper |
| 80 | St. George pluton (A) | Dev. | 085°E, 38°S | This paper |
| 81 | St. George pluton (D) | Dev. | 093°E, 29°N | This paper |
| 82 | St. Stephen pluton (B) | Dev. | 136°E, 43°S | This paper |
| 83 | Bonavista Bay (M) | Dev. | 141°E, 02°N | Murthy and Rao (1979) |
| 84 | Bonavista Bay (R) | Dev. | 080°E, 54°S | Murthy and Rao (1979) |
| 85 | Mount Peyton (granite) | Dev. | 125°E, 63°S | Lapointe (1979) |
| 86 | Catskill red beds | Dev. | 117°E, 47°N | Kent and Opdyke (1978) |

*Groups 1 and 2 are recalculated means from Séguin (1977, Table III), using 9A, 9B, 10A, 10B, 10C, 10D and 13A, 13B, 13C respectively; the $\alpha_{95}$ of the mean directions are given in parentheses. Because of the small number of samples, these poles should be regarded as rough indicators only. WRH and WRB indicate that the poles were obtained from the directions with respect to horizontal and with respect to bedding respectively.


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